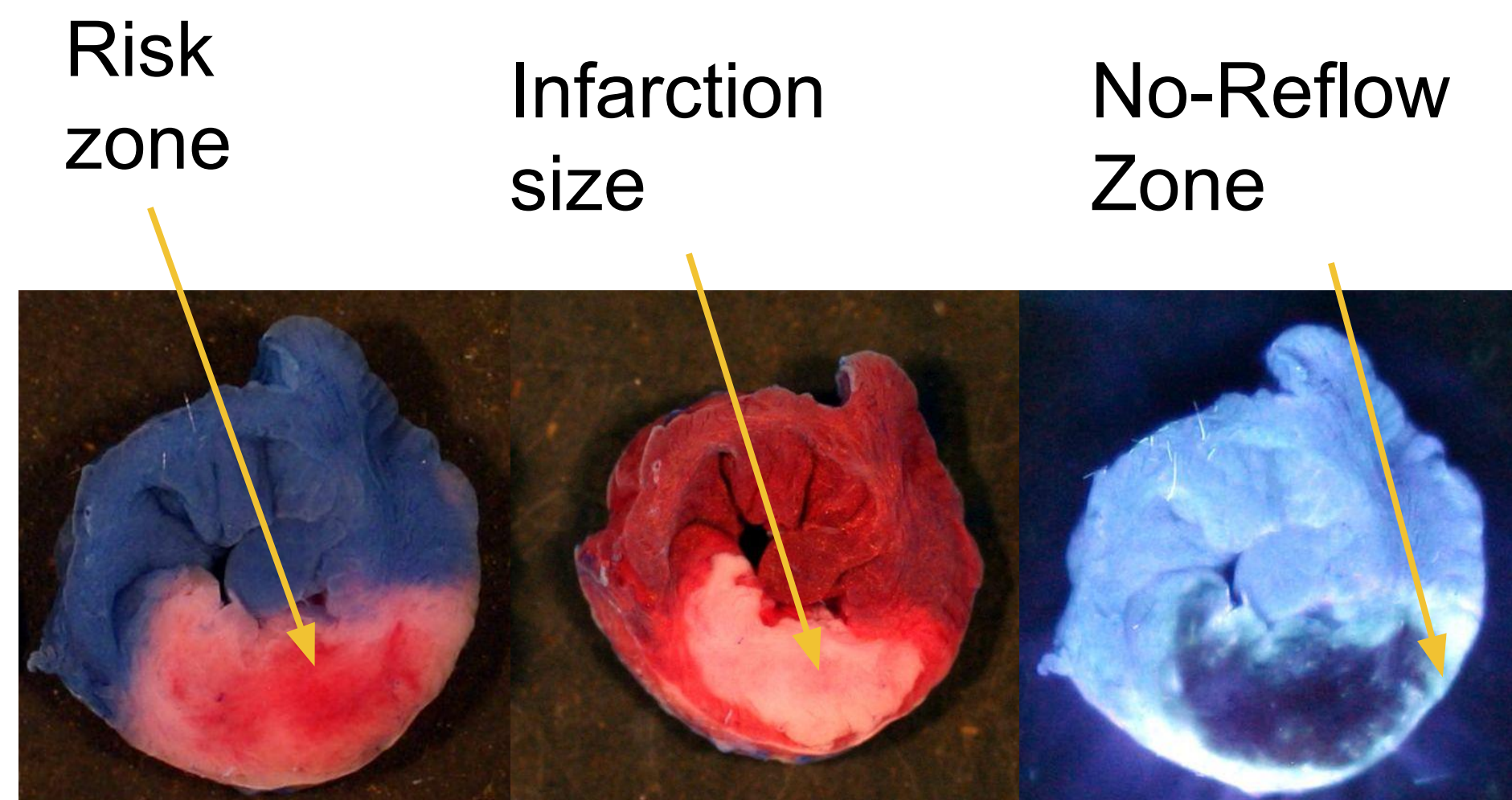


Introduction

We propose a new approach toward fully automated ways to **analyze the mass ratio between the specific target area (Risk zone, infarction size, no-reflow zone) and the total area of the left ventricle of the heart**. Our approach design uses deep learning-based image segmentation (dividing the image into distinct regions that shares the same attributes). This segmentation allows us to compare the target areas with the total area and calculate the mass ratio.

Background

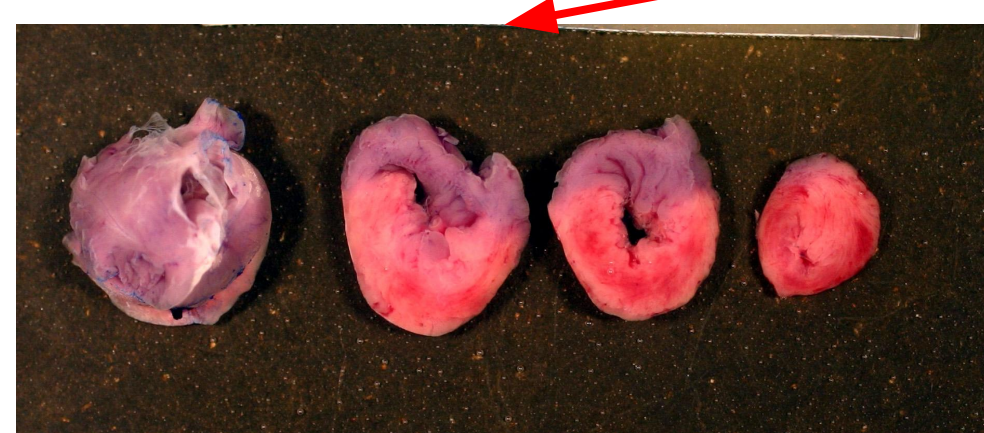
- **Risk zone** is the area at risk of becoming necrotic (cell death) due to inadequate blood supply during an ischemic event.
- **Infarction size** is the area of necrosis in the heart following an interruption of blood supply due to a blockage.
- Within infarction size, **no-reflow phenomenon** occurs. No-reflow zone refers to the areas of the heart that do not receive adequate blood flow after the reopening of previously blocked arteries.



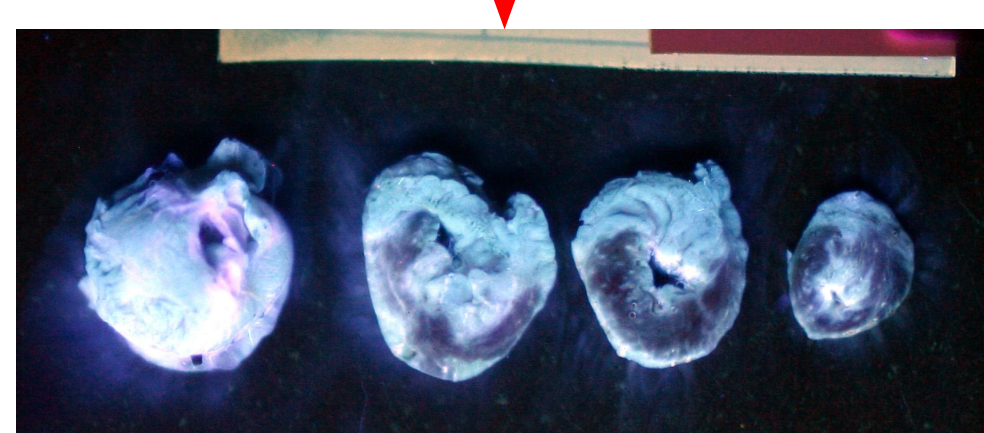
Representative sections of rat heart (left ventricle) showing **risk zone** (left image: red region), **infarction size** (middle image: pale (white) region, the necrotic tissues stained by TTC), and **no-reflow zone** (right image under UV light: dark/non-fluorescent area)

Data Preparation

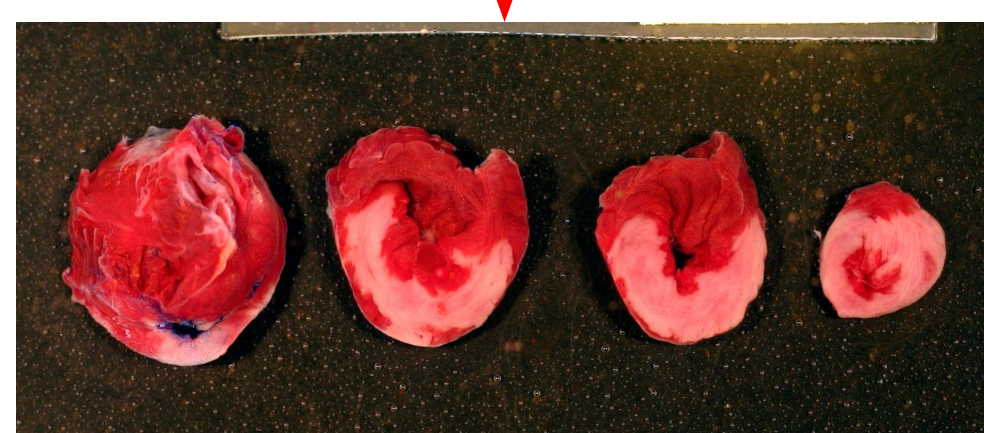
After removal of the right ventricle and the major vessels, the left ventricle was transversely sliced into four sections and photographed.



Slices were weighed, photographed, and stained with blue pigment to determine the risk zone (parts not dyed blue).

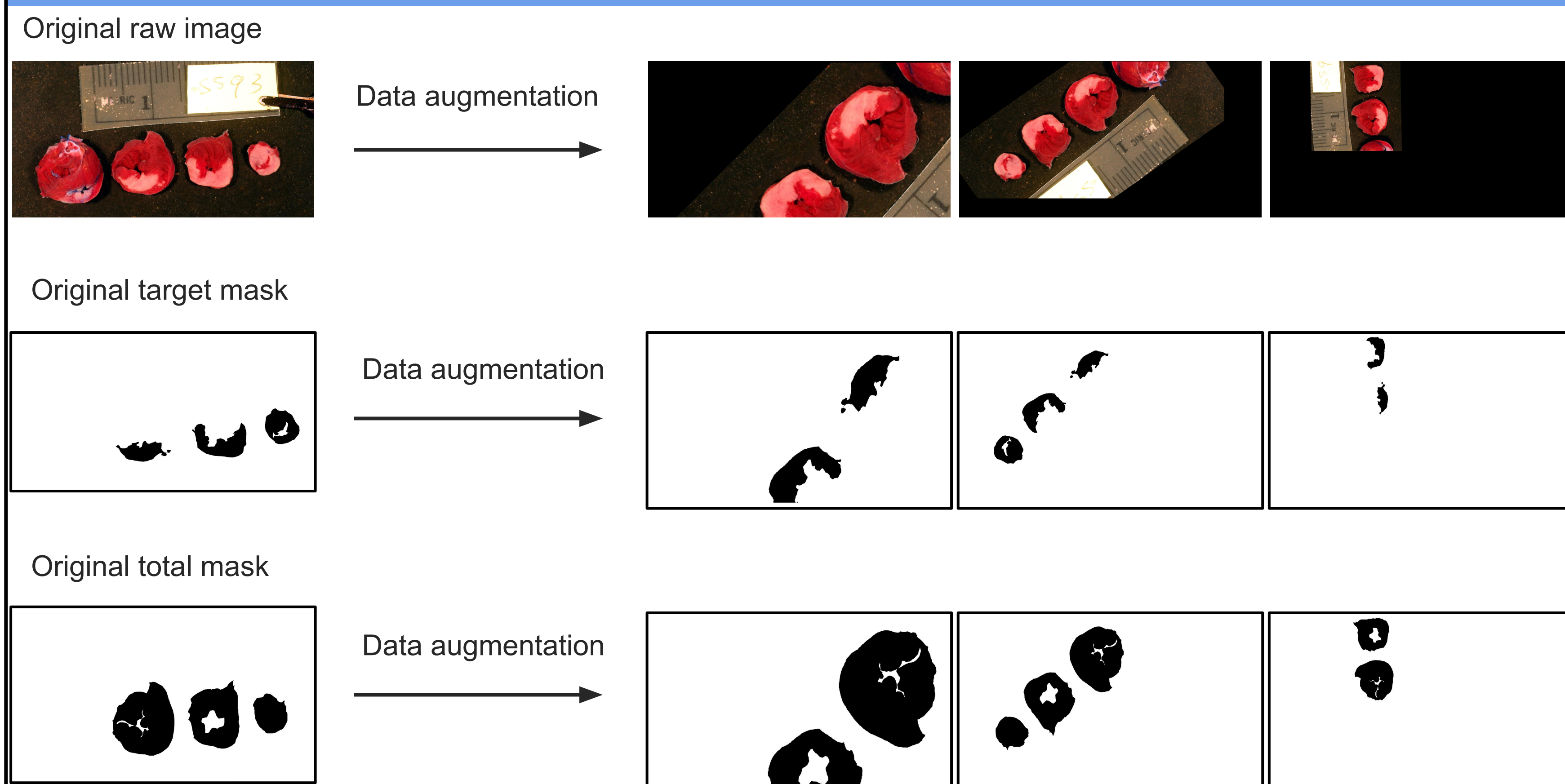


The slices were then photographed under UV light to delineate thioflavin S stained and unstained (no-reflow) regions.



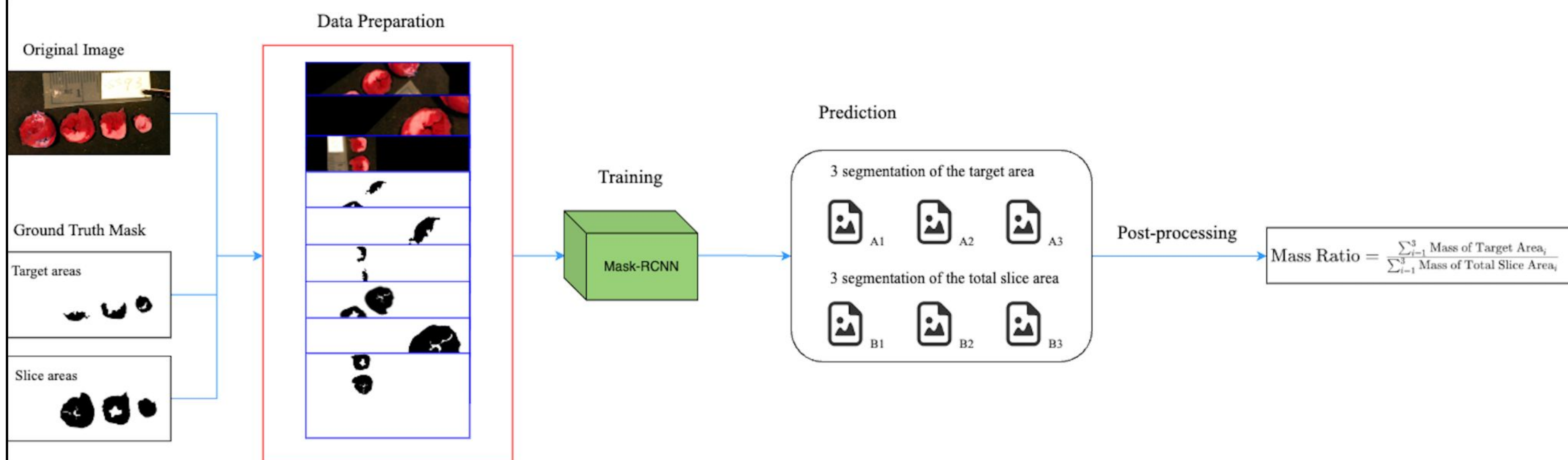
They were then immersed in a 1% solution of triphenyl tetrazolium chloride (TTC), a substance that stains viable cells brick red while rendering dead or necrotic cells white.

Data Augmentation



Given the limited dataset size for this project, we expanded it through data augmentation techniques, including rotation, shifting, and resizing of the original images & masks.

Machine Learning Model Design



Results

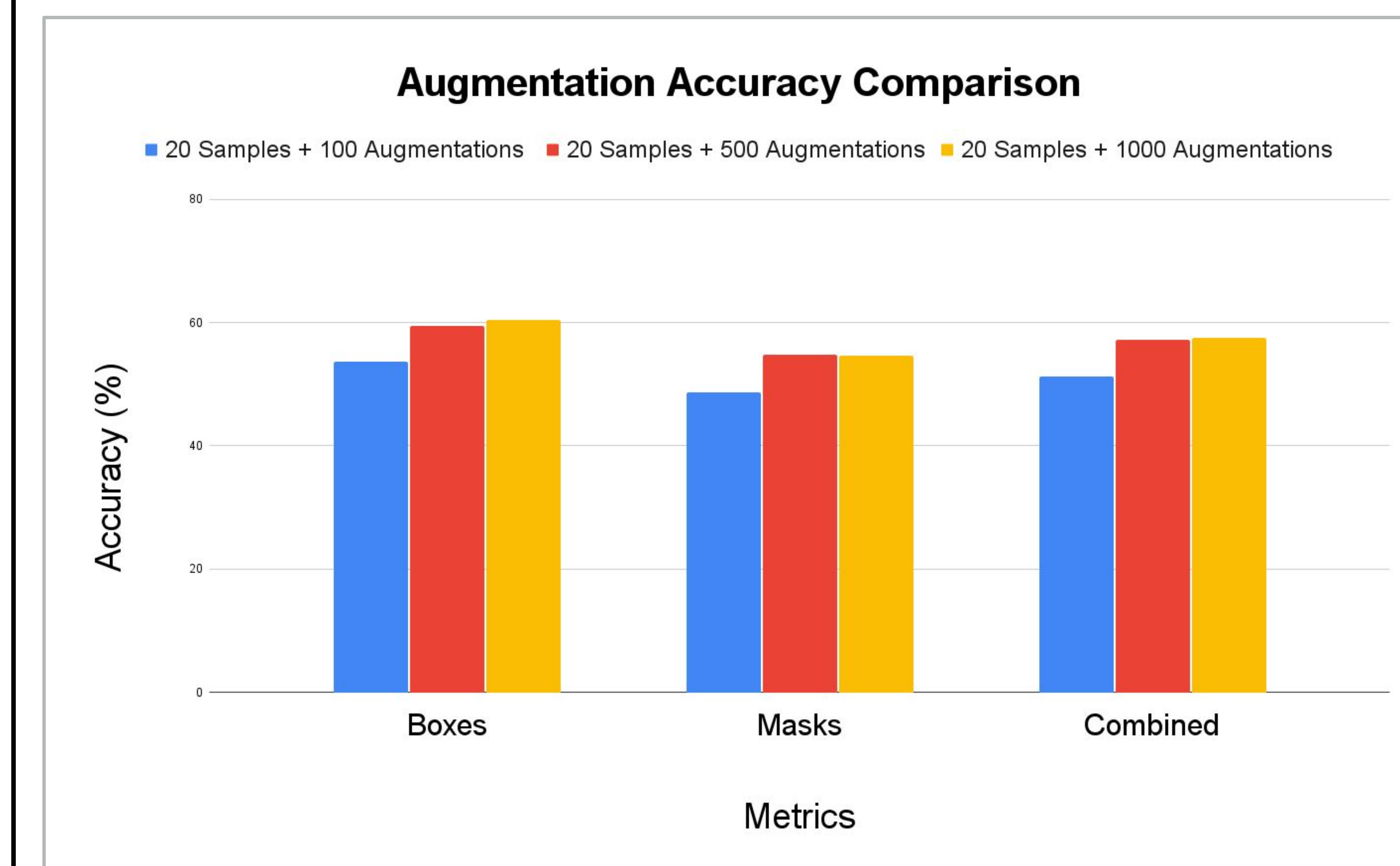


Figure 1:

Accuracy comparison of the model trained with 20 original samples and 100, 500, and 1,000 augmentations.

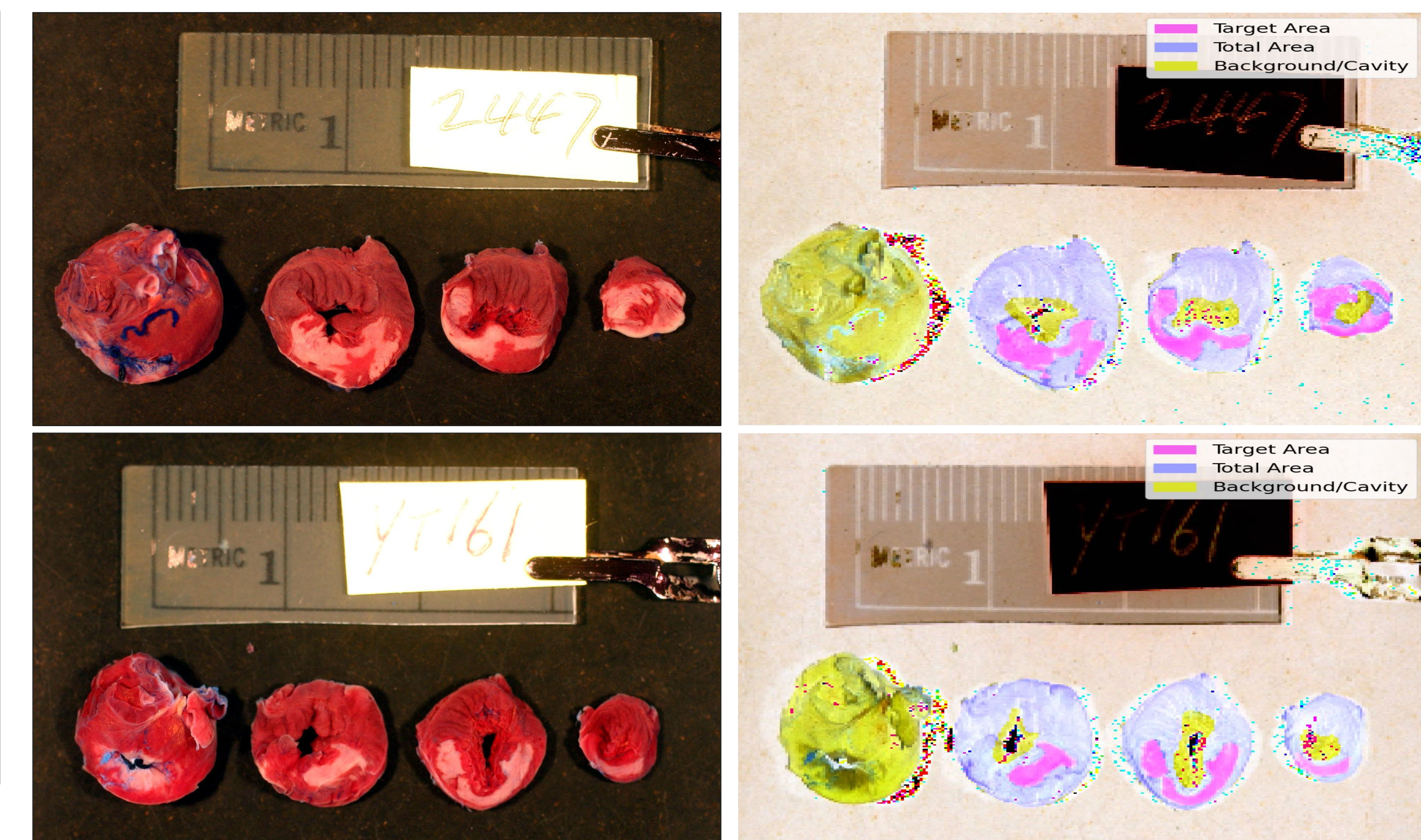


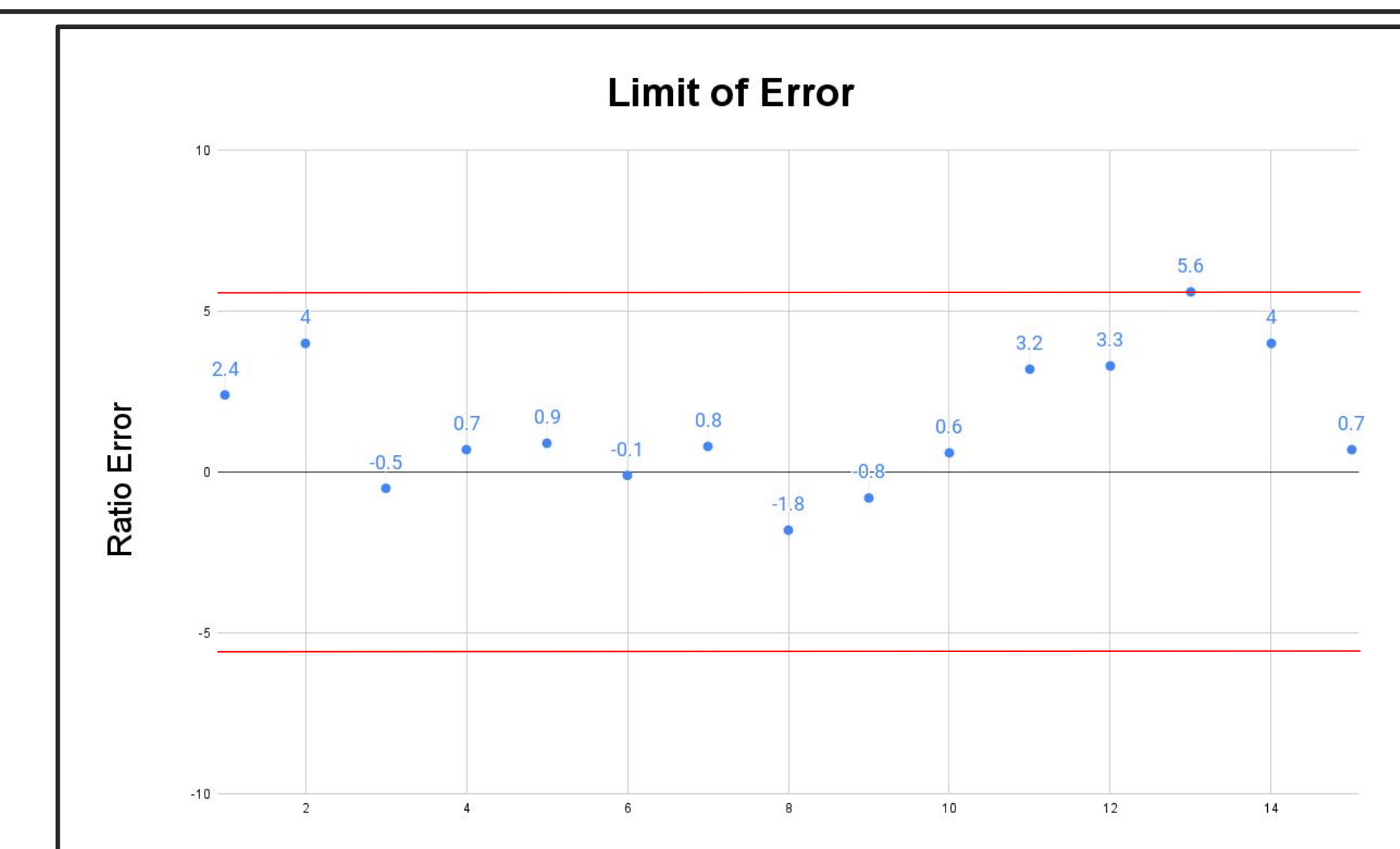
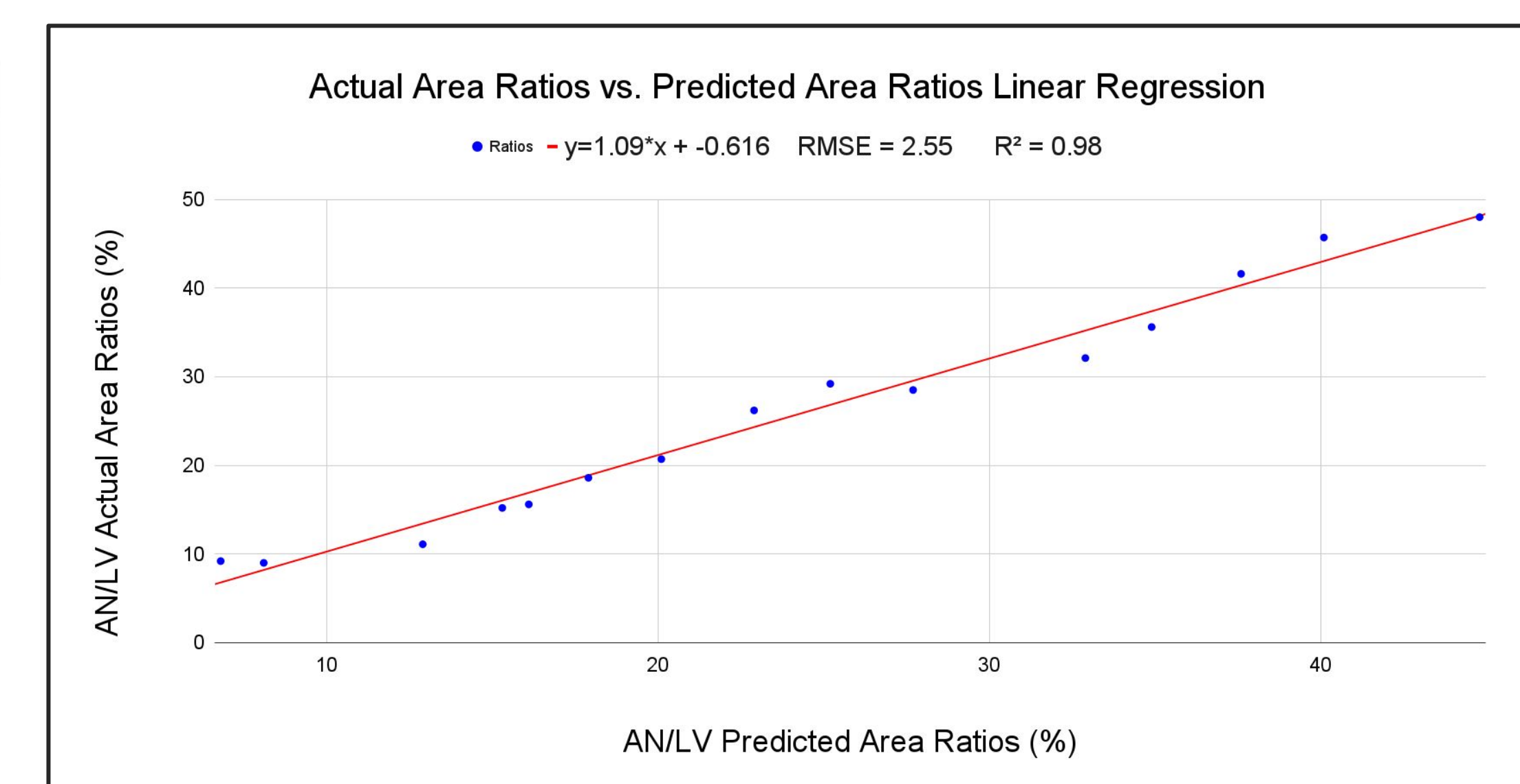
Figure 2:

Original infarction size vs predicted infarction size

Model With the Highest Accuracy (Trained on 100 Original Samples with 1000 Augmentations)

Test Phase Accuracy Summary	
Data	Accuracy (%)
Boxes	62.64
Masks	57.71
Combined	60.18

Test Phase Ratio Predictions Summary		
Images	AN/LV Predicted Area Ratios (%)	AN/LV Actual Area Ratios (%)
IMG_5911.png	6.8	9.2
IMG_5999.png	37.6	41.6
IMG_6028.png	16.1	15.6
IMG_6047.png	17.9	18.4
IMG_6070.png	8.1	9.0
IMG_6492.png	15.3	15.2
IMG_6498.png	27.7	28.5
IMG_6513.png	12.9	11.1
IMG_6526.png	32.9	32.1
IMG_6538.png	20.1	20.7
IMG_6979.png	44.8	48.0
IMG_7004.png	22.9	26.2
IMG_7016.png	40.1	45.7
IMG_7022.png	25.2	29.2
IMG_7028.png	34.9	35.6



Discussion

- The number of original image & mask datasets used in training **significantly impacts** the Mask-RCNN model's accuracy.
- Augmenting the dataset **improves accuracy**, but the gains **slow down** as more augmentations are added.
- The **most accurate model** (trained with 100 samples + 1,000 augmentations) has a maximum ratio deviation of **±5.6%**, a root mean squared error (RMSE) of **2.55** and a R^2 value of **0.98** when comparing predicted area ratios to actual area ratios.

Future Direction

- Develop post processing calculation
- Expand the original image & mask dataset
- Train the model with myocardial risk zone and no-reflow zones.
- Deploy software incorporating the trained machine learning model.

Acknowledgement

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